13. Comparative Analysis of NB-IoT and LTE-M for an IoT Sensor Application

MATLAB CODE:

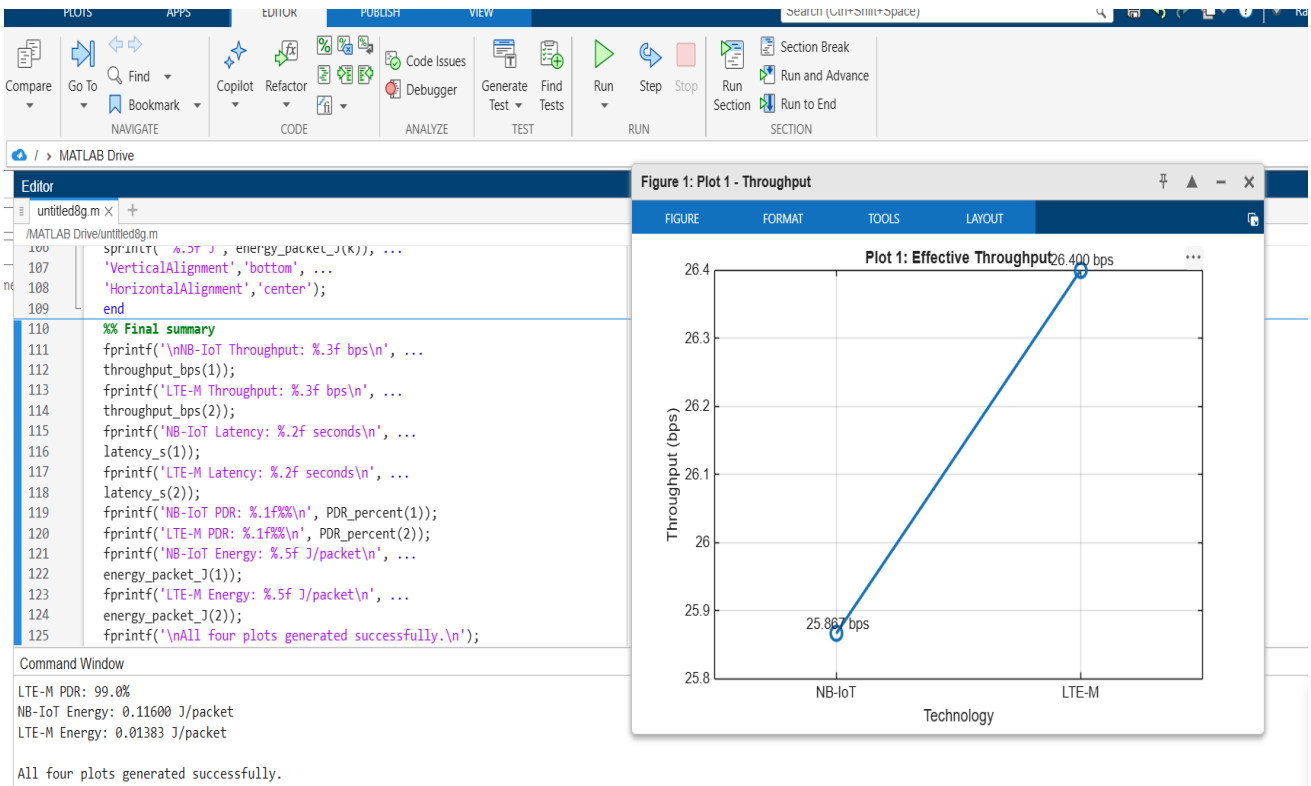
```
clc;
clear;
close all;
%% NB-IoT vs LTE-M Comparative Analysis
% Common sensor parameters
payload_bytes = 200;
sample_interval = 60;
simulation_time = 3600;
% Technology parameters
tech = {'NB-IoT', 'LTE-M'};
data_rate_bps = [20000, 300000];
success_prob = [0.97, 0.99];
latency_s = [2.0, 0.15];
tx_power_W = [0.20, 0.25];
overhead_s = [0.50, 0.05];
%% Calculations
N_tx = floor(simulation_time / sample_interval);
payload_bits = payload_bytes * 8;
expected_rx = N_tx .* success_prob;
PDR_percent = success_prob .* 100;
throughput_bps = ...
expected_rx .* payload_bits / simulation_time;
payload_time_s = payload_bits ./ data_rate_bps;
energy_packet_J = tx_power_W .* ...
(payload_time_s + overhead_s);
energy_bit_J = energy_packet_J ./ ...
(payload_bits .* success_prob);
```

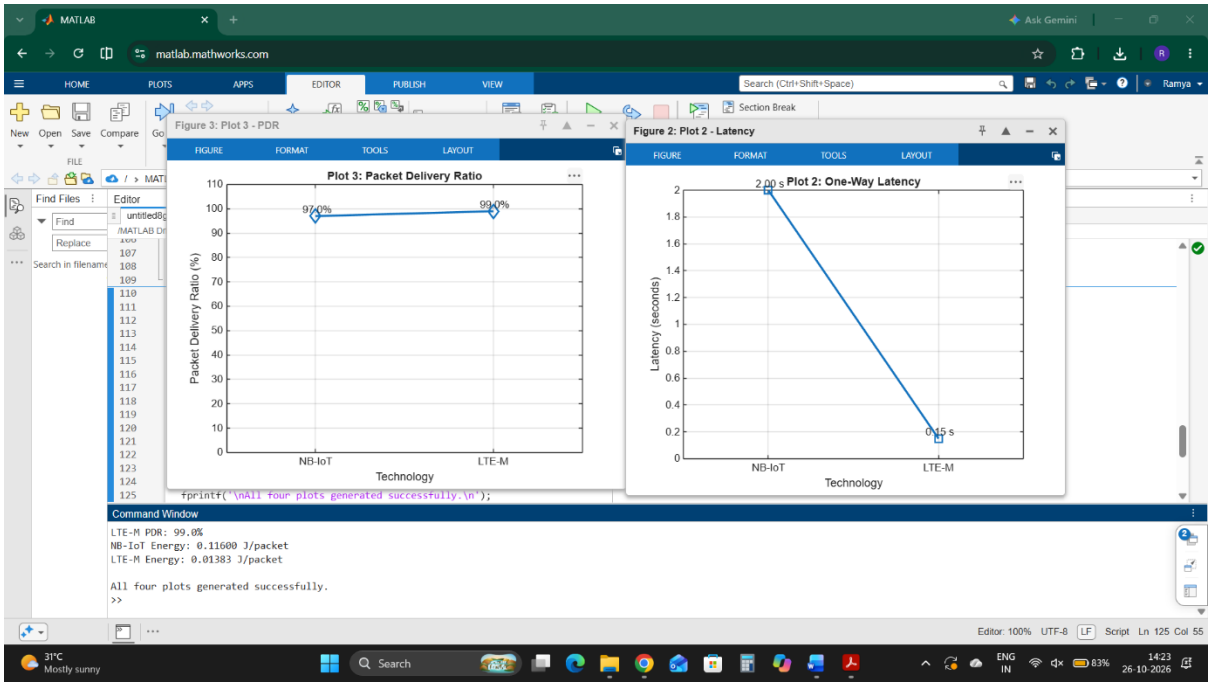
```
%% Display results table
Technology = tech';
GeneratedPackets = [N_tx; N_tx];
ExpectedDelivered = expected_rx';
PDR_percent = PDR_percent';
Throughput_bps = throughput_bps';
Latency_s = latency_s';
EnergyPerPacket_J = energy_packet_J';
EnergyPerDeliveredBit_J = energy_bit_J';
Results = table(Technology, GeneratedPackets, ...
ExpectedDelivered, PDR_percent, Throughput_bps, ...
Latency_s, EnergyPerPacket_J, ...
EnergyPerDeliveredBit_J);
disp('================================================');
disp(' COMPARATIVE ANALYSIS RESULTS');
disp('================================================');
disp(Results);
%% Plot 1: Effective Throughput
figure('Name','Plot 1 - Throughput','Color','w');
plot(1:2, throughput_bps, 'o-', ...
'LineWidth', 2, 'MarkerSize', 9);
set(gca, 'XTick', 1:2, 'XTickLabel', tech);
xlim([0.5 2.5]);
ylabel('Throughput (bps)');
xlabel('Technology');
title('Plot 1: Effective Throughput');
grid on;
for k = 1:2
text(k, throughput_bps(k), ...
```

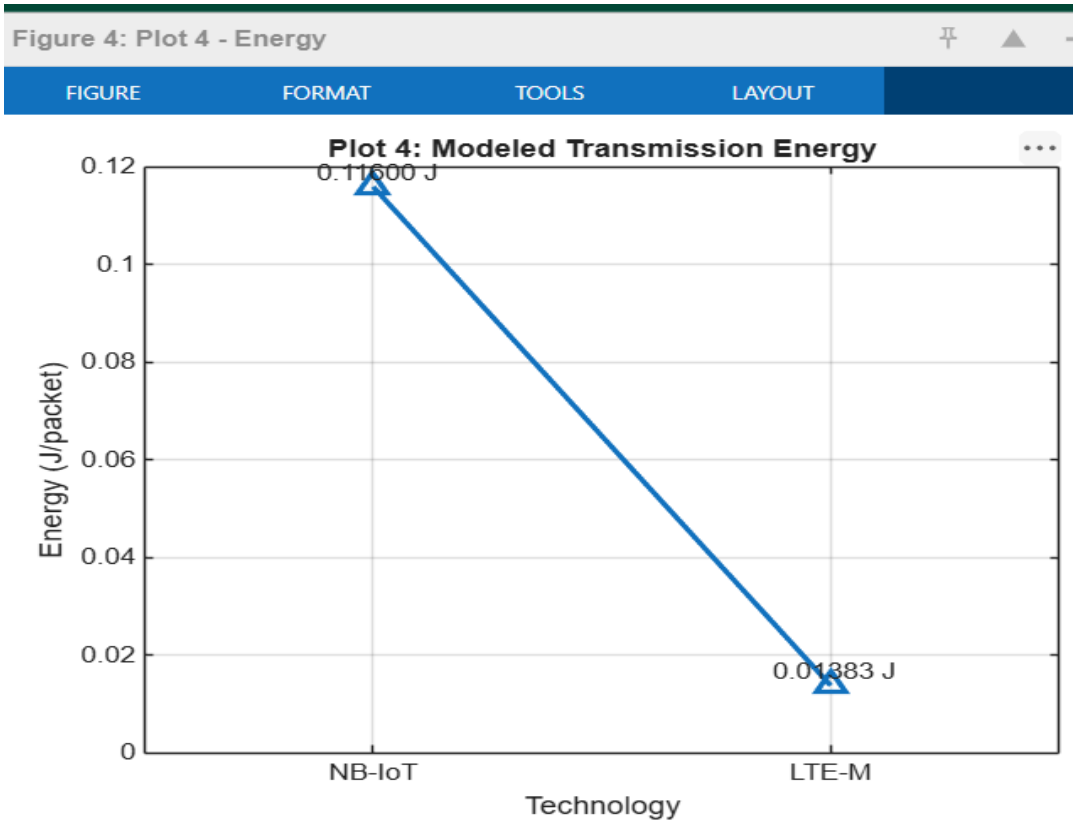
```
sprintf(' %.3f bps', throughput_bps(k)), ...
'VerticalAlignment','bottom', ...
'HorizontalAlignment','center');
end
%% Plot 2: One-Way Latency
figure('Name','Plot 2 - Latency','Color','w');
plot(1:2, latency_s, 's-', ...
'LineWidth', 2, 'MarkerSize', 9);
set(gca, 'XTick', 1:2, 'XTickLabel', tech);
xlim([0.5 2.5]);
ylabel('Latency (seconds)');
xlabel('Technology');
title('Plot 2: One-Way Latency');
grid on;
for k = 1:2
text(k, latency_s(k), ...
sprintf(' %.2f s', latency_s(k)), ...
'VerticalAlignment','bottom', ...
'HorizontalAlignment','center');
end
%% Plot 3: Packet Delivery Ratio
figure('Name','Plot 3 - PDR','Color','w');
plot(1:2, PDR_percent, 'd-', ...
'LineWidth', 2, 'MarkerSize', 9);
set(gca, 'XTick', 1:2, 'XTickLabel', tech);
xlim([0.5 2.5]);
ylim([0 110]);
ylabel('Packet Delivery Ratio (%)');
xlabel('Technology');
```

```
title('Plot 3: Packet Delivery Ratio');
grid on;
for k = 1:2
text(k, PDR_percent(k), ...
sprintf(' %.1f%%', PDR_percent(k)), ...
'VerticalAlignment','bottom', ...
'HorizontalAlignment','center');
end
%% Plot 4: Modeled Transmission Energy
figure('Name','Plot 4 - Energy','Color','w');
plot(1:2, energy_packet_J, '^-', ...
'LineWidth', 2, 'MarkerSize', 9);
set(gca, 'XTick', 1:2, 'XTickLabel', tech);
xlim([0.5 2.5]);
ylabel('Energy (J/packet)');
xlabel('Technology');
title('Plot 4: Modeled Transmission Energy');
grid on;
for k = 1:2
text(k, energy_packet_J(k), ...
sprintf(' %.5f J', energy_packet_J(k)), ...
'VerticalAlignment','bottom', ...
'HorizontalAlignment','center');
end
%% Final summary
fprintf('\nNB-IoT Throughput: %.3f bps\n', ...
throughput_bps(1));
fprintf('LTE-M Throughput: %.3f bps\n', ...
throughput_bps(2));
```

```
fprintf('NB-IoT Latency: %.2f seconds\n', ...
latency_s(1));
fprintf('LTE-M Latency: %.2f seconds\n', ...
latency_s(2));
fprintf('NB-IoT PDR: %.1f%%\n', PDR_percent(1));
fprintf('LTE-M PDR: %.1f%%\n', PDR_percent(2));
fprintf('NB-IoT Energy: %.5f J/packet\n', ...
energy_packet_J(1));
fprintf('LTE-M Energy: %.5f J/packet\n', ...
energy_packet_J(2));
fprintf('\nAll four plots generated successfully.\n');
```




Figure 3: Plot 3 - PDR
Plot 3: Packet Delivery Ratio
97.0%
99.0%
Packet Delivery Ratio (%)
NB-IoT
LTE-M
Technology
Figure 2: Plot 2 - Latency
Plot 2: One-Way Latency
2.00 s
0.15 s
Latency (seconds)
NB-IoT
LTE-M
Technology
Command Window
LTE-M PDR: 99.0%
NB-IoT Energy: 0.11600 J/packet
LTE-M Energy: 0.01383 J/packet
All four plots generated successfully.


Figure 4: Plot 4 - Energy
FIGURE
FORMAT
TOOLS
LAYOUT
Plot 4: Modeled Transmission Energy
0.11600 J
0.01383 J
Energy (J/packet)
NB-IoT
LTE-M
Technology